Thesis Defense

Predicting Lung Cancer Incidence from CT Imagery

Vu Hung Son

Department of Information and Communication Technology
University of Science and Technology of Hanoi

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Overview

1. Introduction
2. Theoretical Background
3. Material and Methodology
4. Result and Discussion
5. Conclusion and Future Work

Background and Context

- **Lung Cancer Statistics:**
  - 2.2M new cases and 1.8M deaths globally in 2020 (WHO)
  - High incidence in Vietnam, especially among men
- **Importance of Early Detection:**
  - Early diagnosis boosts survival rates
  - CT scans are effective but require skilled analysis, which is time-consuming
- **AI in Lung Cancer Detection:**
  - Automate detection, reducing doctors' workload
  - Improve accuracy and speed in identifying small cancerous lesions
- **Project Aim:**
  - Develop a model for CT scan analysis
  - Assist doctors with faster, more accurate lung cancer detection

Problem Statement and Objectives

Problem Statement

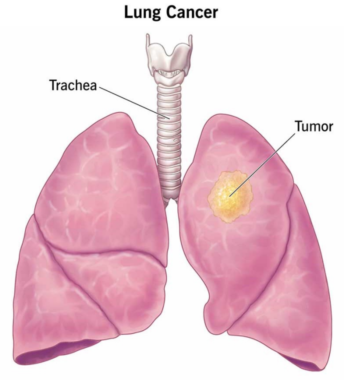
- **Traditional Methods Limitations:** Accuracy depends on the doctor, time-consuming, difficult to detect early, exposure to radiation
- **Deep Learning Approach:** Offers high accuracy, automation, early detection, decision support, and personalization
- **Conclusion:** Transitioning to deep learning can address traditional limitations while offering advantages

Research Objectives

- Explore the potential of CNN model in complex medical tasks, particularly in analyzing medical imaging data
- Develop a framework for processing and analyzing 3D medical data, enhancing CNN capabilities

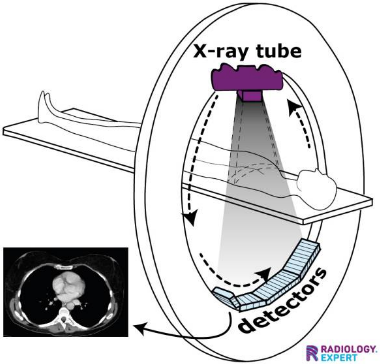
Lung Cancer Overview

- **Lung Cancer:** Abnormal cell growth in lungs, tumor formation, potential metastasis; high incidence and mortality rates
- **Risk Factors:** Tobacco, secondhand smoke, air pollution, radon, asbestos, family history
- **Conclusion:** Leading cause of cancer deaths; early detection crucial for improved survival



Computed Tomography Scan Overview

- **CT Scan:** Advanced imaging technique using X-rays and computer processing for detailed lung images
- **Applications:** Detects small lung nodules, identifies tumor size and location, assesses cancer spread
- **Benefits:** Early detection, accurate diagnosis, treatment monitoring, crucial for improving lung cancer outcomes



Application in Medical Imaging

Deep Learning

- Advanced models analyze complex patterns in medical images (X-rays, CT scans, MRI)
- Detect subtle changes, offering faster, more accurate diagnoses

3D CNNs

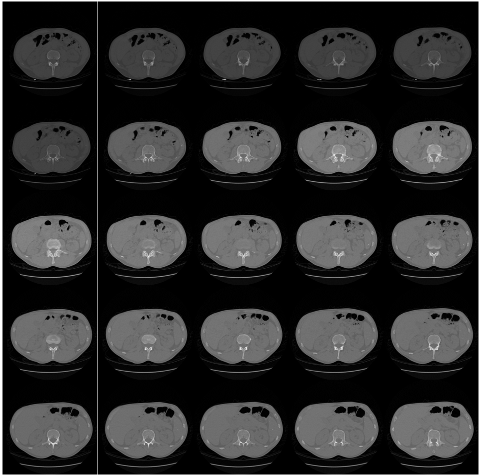
- Analyzes 3D medical data like CT and MRI scans
- Effective for tumor detection, image segmentation, and structural analysis

Benefits

- Improves detection accuracy and aids in disease diagnosis
- Supports personalized treatment and reduces doctors' workload

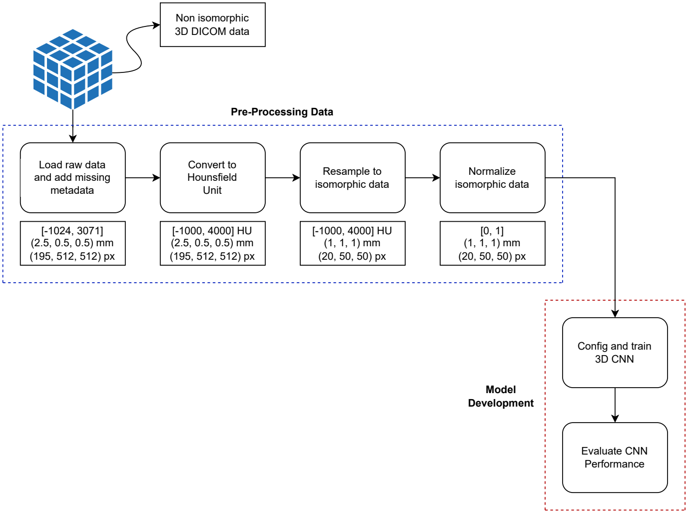
Material

- **Dataset:** Utilized a subset of the NLST dataset with 769 low-dose CT scans for lung cancer diagnosis
- **Characteristics:** Imbalanced data: 574 non-cancer and 195 cancer cases; variations in the number of CT slices per patient
- **Challenges:** Non-isomorphic and imbalance data impacted training of the 3D CNN model, affecting performance



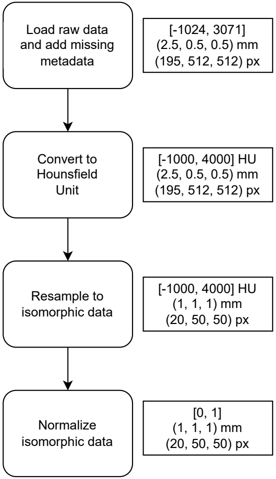
Overall Framework

- **Step 1:** EDA
- **Step 2:** Prepare data
- **Step 3:** Train model



Data Preprocessing

- **Data Cleaning:** Arranged DICOM files, calculated slice thickness
- **HU Conversion:** Standardized tissue densities using Hounsfield Units
- **Resampling:** Uniform voxel size (1x1x1) mm, limited to 20 slices, resized to 50x50 pixels
- **Normalization:** Min-max normalization, zero-centering for consistent model training



Model Development

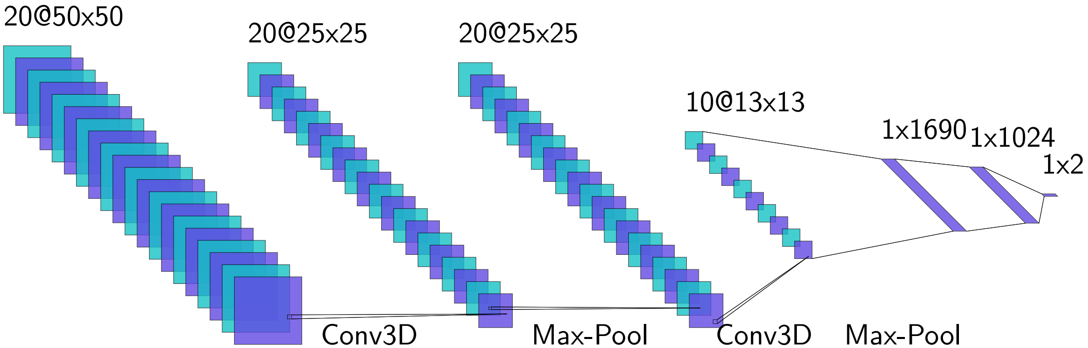


Figure: 3D CNN Architecture

Model Development

- **Convolutional:** 2 layers with 32, 64 filters, kernel size (3,3,3), ReLU activation, 'same' padding
- **Max-Pooling:** Pool size (2,2,2) for down-sampling, reduces feature map size
- **Flatten:** Converts 3D features to 1D for fully connected layers
- **Fully Connected:** Dense layer with 1024 neurons, ReLU activation, dropout rate 0.2
- **Output Layer:** Softmax activation for binary classification (cancer vs. non-cancer)

- **Image Size:** 50x50 pixels
- **Number of Slices:** 20 slices
- **Batch Size:** 10
- **Epochs:** 10
- **Learning Rate:** 1e-3

Result and Discussion

Metric	Value
Accuracy	0.68
Sensitivity	0.0
Specificity	1.0
Precision	1.0
AUC ROC	0.61

Table: Model Performance

- Accuracy: 68% misleading; favors non-cancer predictions
- Sensitivity: 0.0; failed to detect cancer cases
- Specificity/ Precision: Both 1.0; not informative without cancer detection
- Challenges: Highlights difficulties in lung cancer detection; need for model improvements

Conclusion and Future Work

Conclusion

- **Model Performance:** Bias towards non-cancer cases, poor sensitivity
- **Challenges:** Data imbalance, low signal-to-noise ratio, limited dataset

Future Work

- **Model Improvement:** Test advanced architectures (e.g., 3D ResNet)
- **Dataset:** Expand and balance data, use augmentation
- **Applications:** Classify stages, tumor segmentation, support clinical decisions

Q&A

Thank you for your attention!

If you have questions, please feel free to ask